



Final – MATH 518 – Winter 2025

DATE: Tuesday, December 9 TIME: 2:00PM – 5:00PM

EXAMINER: Romain Branchereau

ASSOCIATE EXAMINER: Eyal Goren

First name (please write as legibly as possible within the boxes)

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Last name

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Student ID number

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- There are 6 problems, for a total of 70 points.
- Notes, calculators and other electronic devices are not permitted.
- Throughout the exam, the field k is of characteristic 0 and algebraically closed.
- The last pages can be used as scrap paper and will not be graded.



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1. (20 points) For each of the following statements, determine if it is true or false. Give a brief justification (short proof or counterexample).
 - (a) For certain integers $n \geq 1$, the quasi-projective variety $X = k^n \setminus \{0\}$ is affine.
 - (b) The variety $\mathbb{P}^1(k) \times \mathbb{P}^1(k)$ is neither affine nor projective.
 - (c) If $X = V(f, g) \subseteq k^3$ is an affine variety given by two nonzero polynomials $f, g \in k[x, y, z]$ of positive degree, then $\dim(X) = 1$.
 - (d) If an affine variety X is connected (in the Euclidean topology), then it is irreducible in the Zariski topology.
 - (e) Any two distinct lines in $\mathbb{P}^2(k)$ intersect in a single point.



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2. (10 points) Find the irreducible components of

$$X = V(x^3 + yz, x(z + y), x^3 - z^2) \subseteq k^3.$$

Determine the dimension of each irreducible component.



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3. (10 points) Let $f, g: X \longrightarrow Y$ be morphisms between two quasi-projective varieties X and Y . If f and g agree on a nonempty open set U , show that they agree on X . *Show it for affine varieties first.*



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4. (10 points) Consider the blowup map $\phi: k^2 \longrightarrow k^2$ defined by $\phi(x, y) = (x, xy)$. Determine the image of ϕ and show that it is a dense subset that is neither open nor closed.



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5. (10 points) Let $\phi: \mathbb{P}^1(k) \longrightarrow \mathbb{P}^2(k)$ be the map defined by

$$\phi([u: v]) = [u^2 : u^2 - uv : v^2].$$

Let $X = V(f)$ be the image of ϕ . Determine f and show that ϕ an isomorphism onto X .



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6. (10 points) Let $X \subseteq k^2$ be the curve $X = V(f)$ where $f = y^2 - x^2(1 + x)$. Determine the points where X is smooth, and use blowups to find a smooth curve Y in k^2 that is birational to Y .



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